

THIS IS UPPERCASE

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Abstract

This thesis explores the arithmetic and geometric properties of torsion points on elliptic curves, with a specific focus on curves admitting complex multiplication by the ring of Gaussian integers, $\mathbb{Z}[i]$. We begin by establishing the foundational algebraic geometry of elliptic curves, utilizing the Riemann-Roch theorem to derive the general and short-form Weierstrass equations. After defining the geometric group law and the analytic isomorphism between the complex torus $\mathbb{C}/\Lambda$ and the curve $E(\mathbb{C})$, we provide a rigorous treatment of the torsion subgroup structure. The core of this work focuses on the construction and factorization of division polynomials. For the specific curve $y^2 = x^3 + x$, we derive explicit multiplication-by- π maps for Gaussian integers $\pi = n + im$. We demonstrate that the composite division polynomial $\Psi_\pi = \psi_\pi \psi_{\bar{\pi}}$ within $\mathbb{Z}[i][x]$. By applying the Eisenstein criterion over the Gaussian integers, we verify the irreducibility of these factors and conclude that the p -torsion subgroup decomposes into distinct eigenspaces when $p \equiv 1 \pmod{4}$.

These theoretical results are further supported by computational verification using a custom symbolic algebra implementation in Python. By applying the Eisenstein criterion over the Gaussian integers we show that the Galois group of the field extension of $\mathbb{Q}(i)$ generated by coordinates of π -torsion points in E is cyclic of order $p - 1$.

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Chapter 1

Preliminaries on Elliptic Curves

Before studying the arithmetic of torsion points of such curves, we must establish their geometric foundations. We begin by deriving the equation of the curve using the Riemann-Roch theorem, then explore the group law of these curves.

1.1 Derivation of the Weierstrass Equation

The central topic of this paper discusses Gaussian torsion points of elliptic curves, it may be helpful to define what we mean by an *elliptic curve*. An elliptic curve E has the typical definition as:

Definition 1.1. An *elliptic curve* is a non-singular projective algebraic curve of genus one, with an identity point $\mathcal{O}$.

Often these curves are encountered as cubic equations in the plane. However, their structure comes from a deeper theory of algebraic curves.

1.1.1 The Riemann-Roch Theorem

The Riemann-Roch theorem is the fundamental link between a curve's topological genus and its algebraic properties. It allows us to determine the dimension of the space of rational functions that have poles of a limited order at specific points.

A divisor on C is a sum of points

$$D = \sum_{P \in C} n_P P,$$

where all but finitely many n_P are zero.

Let C be a non-singular projective curve over an algebraically closed field $\bar{F}$. For a divisor D on C , let $\mathcal{L}(D)$ denote the space of rational functions f such that $(f) + D \geq 0$. The dimension of this space is denoted by $\ell(D)$. The Riemann-Roch theorem relates this dimension to the degree of the divisor and the genus of the curve.

Theorem 1.2 (Riemann-Roch). *Let K be a canonical divisor on C . Then for any divisor D :*

$$\ell(D) - \ell(K - D) = \deg(D) - g + 1 \quad (1.1)$$

Proof. A full proof of this theorem is beyond the scope of this thesis; however, a classical proof can be found in [Ful69], and a generalization in [Har77]. $\square$

For an elliptic curve, by definition, we have genus $g = 1$. Consequently, $\deg(K) = 2g - 2 = 0$. When we consider a divisor $D = n\mathcal{O}$ for $n \geq 1$, we note that $\deg(n\mathcal{O}) = n > 0$, which implies

$\ell(K - n\mathcal{O}) = 0$. Thus, the formula simplifies elegantly to:

$$\ell(n\mathcal{O}) = n + 1 - 1 = n.$$

1.1.2 Deriving the Weierstrass Equation

We fix the point $\mathcal{O} \in E$ to serve as the identity element. We examine the spaces $\mathcal{L}(n\mathcal{O})$, consisting of functions with a pole at $\mathcal{O}$ of order at most n . We use the simplified Riemann-Roch result from before, we construct a basis for functions on the curve: We now construct a basis for functions on the curve:

- $n=1$: $\ell(\mathcal{O}) = 1$. The only functions with no poles (or a pole of order 0) are constants. Basis: $\{1\}$.
- $n = 2$: $\ell(2\mathcal{O}) = 2$. We introduce a function x with a pole of order exactly 2 at $\mathcal{O}$. Basis: $\{1, x\}$.
- $n = 3$: $\ell(3\mathcal{O}) = 3$. We introduce a function y with a pole of order exactly 3 at $\mathcal{O}$. Basis: $\{1, x, y\}$.

The Linear Dependence

Consider the space for $n = 6$. By our formula, $\ell(6\mathcal{O}) = 6$. However, we can form the following seven functions using powers of x and y , all of which have pole orders less than or equal to 6:

$$\{1, x, y, x^2, xy, y^2, x^3\}$$

Since there are 7 vectors in a 6-dimensional space, they must be linearly dependent. Thus, there exist constants a_i such that:

$$y^2 + a_1xy + a_2y = a_3x^3 + a_4x^2 + a_5x + a_6 \quad (1.2)$$

This is the general Weierstrass equation. By rescaling variables (assuming $\text{char } \bar{F} \neq 2, 3$), we obtain the short form:

$$y^2 = x^3 + ax + b \quad (1.3)$$

1.2 Projective Coordinates

The equation (1.3) describes the curve in the *affine plane* $\mathbb{A}^2$. However, elliptic curves are *projective* varieties [Sil09], meaning they must be closed and compact. The affine model is missing the point $\mathcal{O}$ (the pole of x and y).

To include this point, we embed the curve in the *projective plane* $\mathbb{P}^2$ using homogeneous coordinates $[X : Y : Z]$. The relationship to affine coordinates is:

$$x = \frac{X}{Z}, \quad y = \frac{Y}{Z}$$

Here, Z acts as the homogenizing variable. Substituting these into (1.3) yields:

$$\left(\frac{Y}{Z}\right)^2 = \left(\frac{X}{Z}\right)^3 + a\left(\frac{X}{Z}\right) + b$$

Multiplying through by Z^3 to clear denominators gives the homogeneous Weierstrass equation:

$$Y^2Z = X^3 + aXZ^2 + bZ^3 \quad (1.4)$$

1.2.1 The Point at Infinity

The “missing” point $\mathcal{O}$ corresponds to the location where $Z = 0$. Setting $Z = 0$ in the homogeneous equation:

$$Y^2(0) = X^3 \implies X = 0$$

In projective space, coordinates cannot be simultaneously zero, so $Y \neq 0$. We normalize to get the unique point at infinity:

$$\mathcal{O} = [0 : 1 : 0]$$

This is the geometric realization of the abstract point $\mathcal{O}$ used in our Riemann-Roch derivation. This is what we will use as the identity element for all curves in this thesis. The importance of this point is so that this is the only point we must consider as a three dimensional point, all other points in this paper will be x, y points but we must always remember $\mathcal{O}$ because it does not show up in the short Weierstrass equation.

1.3 The Group Law

A feature of elliptic curves is that their points form an abelian group (equivalently a $\mathbb{Z}$ -module and in Chapter 3 we find some curves form $\mathbb{Z}[i]$ -modules). To visualize this, consider the curves $E : y^2 = x^3 - x$ and $C : y^2 = 2x^3 + x$. Notice that C has one root and E has three, these curves will look different on the plane, but their groups remain unaffected. The real loci in Figure 1.1 and Figure 1.2 on the facing page give us examples of what two elliptic curves look like.

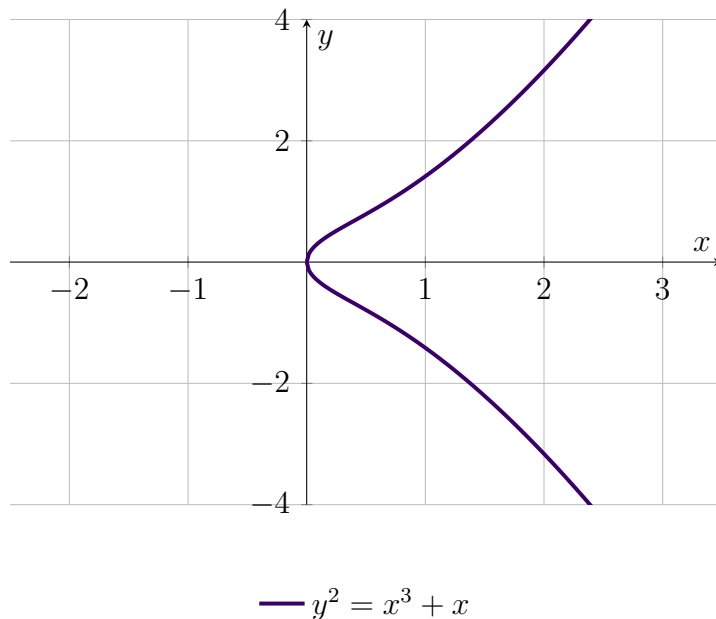


Figure 1.1: Locus of a connected elliptic curve.

1.3.1 Geometric Addition

For two points $P, Q \in E$, the sum $P + Q$ is defined geometrically:

1. Draw the line connecting P and Q (the secant line).
2. Let $P * Q$ be the third point of intersection which counting multiplicity is guaranteed by the fundamental theorem of algebra.
3. Reflect $P * Q$ across the x -axis to obtain $P + Q$.

This process is illustrated in Figure 1.3 on page 6. Notice that even though the curve has two disjoint components (because it has three real roots), the group law functions seamlessly across them.

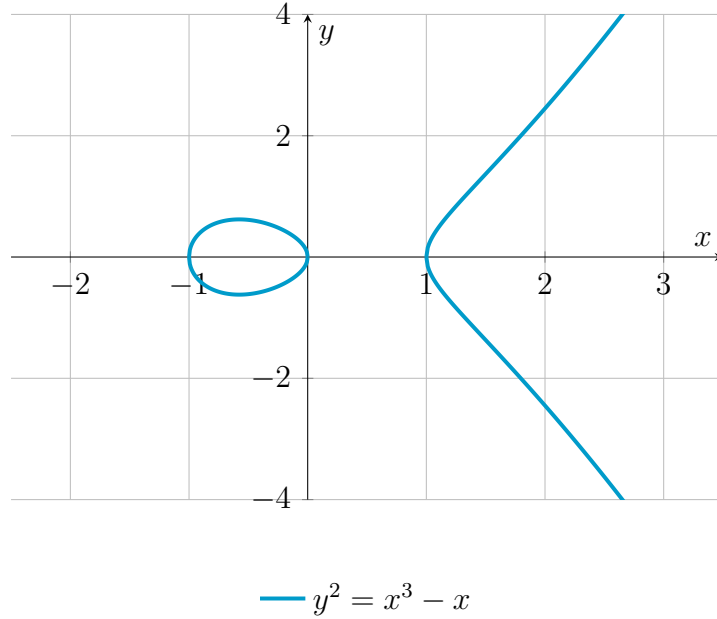


Figure 1.2: Locus of a two component elliptic curve.

Explicitly, on a curve given by the Weierstrass equation $C : y^2 = x^3 + ax + b$, and $P, Q \in C$, if $P = (x_P, y_P)$ and $Q = (x_Q, y_Q)$, the coordinates of $P + Q$ are given by the algebraic formulas:

$$\begin{aligned}\lambda &= \frac{y_Q - y_P}{x_Q - x_P} \quad (P \neq Q), \\ x_{P+Q} &= \lambda^2 - x_P - x_Q, \\ y_{P+Q} &= \lambda(x_P - x_{P+Q}) - y_P.\end{aligned}$$

If $P = Q$ you follow the same procedure as before although since there is no secant line you must use implicit differentiation to find:

$$\lambda = \frac{3x^2 + a}{2y} \quad (P = Q).$$

From this we can define a map $[2] : C \rightarrow C$ which sends $P \mapsto [2]P = P + P$, is called the *point doubling map* and the formula:

$$x([2]P) = \frac{x^4 - 2ax^2 - 8bx + a^2}{4x^3 + 4ax + 4b} \quad (1.5)$$

is called the *point doubling or duplication formula*.

We can find points of order two using this formula. A point of order two is a point P such that $[2]P = \mathcal{O}$. On the curve $y^2 = x^3 - x$ the points of order two are found by finding the roots of $4x^3 + 4x$. Factoring this we find $4x^3 + 4x = 4x(x + 1)(x - 1)$. Thus the x coordinates are

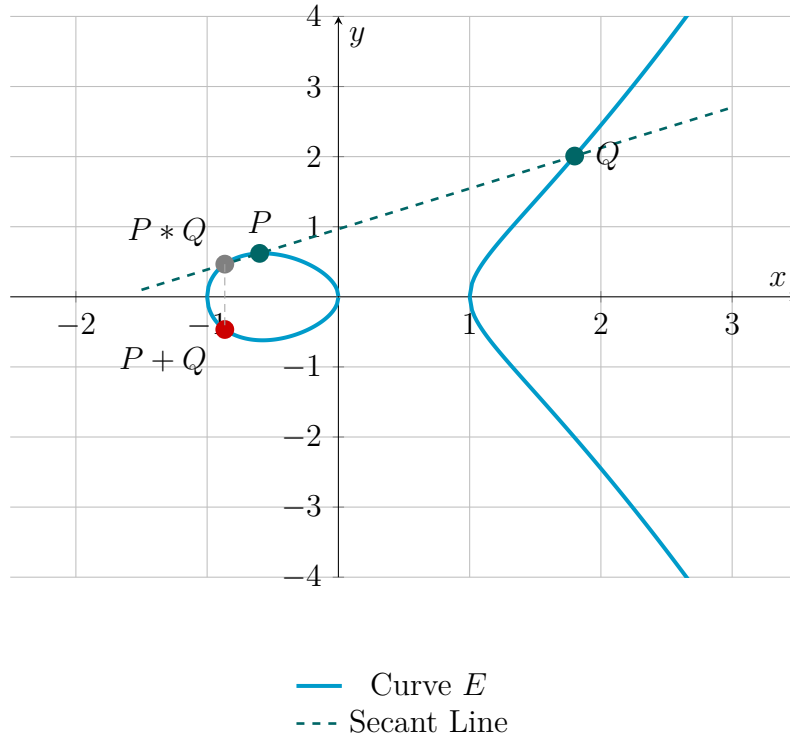


Figure 1.3: Group addition on a curve with three roots.

$x = -1, 0, 1$. Plugging into the curve equation we find that $y = 0$ for each x . Additionally, $[2]\mathcal{O} = \mathcal{O}$, Thus we have found all two torsion points of the curve C ,

$$C[2] = \{\mathcal{O}, (-1, 0), (0, 0), (1, 0)\}.$$

Chapter 2

Division Polynomials and Torsion Structure

Having established the geometric group law, we now turn to an algebraic characterization of the torsion points. To do this, we require a sequence of polynomials that vanish precisely at the points of order n . These are the division polynomials.

2.1 Division Polynomials

Let $\bar{F}$ be a field of characteristic different from 2 and 3. Let C be an elliptic curve given by the Weierstrass equation

$$C : y^2 = x^3 + bx + c, \quad b, c \in \bar{F}.$$

In this section, we construct the division polynomials ψ_n . Division polynomials are defined recursively and they correspond to multiplication-by- n maps.

Definition 2.1. The *division polynomials* $\psi_n \in \mathbb{Z}[x, y]$ are defined recursively as follows:

Base Cases:

$$\psi_1 = 1,$$

$$\psi_2 = 2y,$$

$$\psi_3 = 3x^4 + 6bx^2 + 12cx - b^2,$$

$$\psi_4 = 4y(x^6 + 5bx^4 + 20cx^3 - 5b^2x^2 - 4bcx - 8c^2 - b^3).$$

Inductive Steps ($n \geq 2$):

$$\psi_{2n+1} = \psi_{n+2}\psi_n^3 - \psi_{n-1}\psi_{n+1}^3, \tag{2.1}$$

$$2y\psi_{2n} = \psi_n (\psi_{n+2}\psi_{n-1}^2 - \psi_{n-2}\psi_{n+1}^2). \tag{2.2}$$

To ensure the map is well-defined, we must verify that the recursive division by $2y$ yields a polynomial.

Proposition 2.2. *For all $m \geq 1$, $\psi_m \in \mathbb{Z}[x, y]$. Furthermore, considering ψ_m as a polynomial in variables x and y :*

1. *If m is odd, then $\psi_m \in \mathbb{Z}[x]$.*
2. *If m is even, then $\psi_m \in 2y \cdot \mathbb{Z}[x]$.*

Proof. We proceed by induction on m . The base cases $m = 1, 2, 3, 4$ satisfy the proposition by inspection. Assume the proposition holds for all integers $k < m$.

We first examine the recurrence for when m is even, letting $m = 2n$:

$$\psi_{2n} = \frac{\psi_n}{2y} (\psi_{n+2}\psi_{n-1}^2 - \psi_{n-2}\psi_{n+1}^2).$$

- **Case 1.1: m even, n even.** By the inductive assumption, ψ_n is a multiple of $2y$, so $\psi_n/2y \in \mathbb{Z}[x]$. Since n is even, $n \pm 2$ are even and $n \pm 1$ are odd. The expression in parentheses is of the form $2y \cdot f_1(x) - 2y \cdot f_2(x)$, thus $\psi_{2n} \in 2y\mathbb{Z}[x]$.
- **Case 1.2: m even, n odd.** Since n is odd, $\psi_n \in \mathbb{Z}[x]$. However, $n \pm 1$ are even, so they contain $2y$. The term in parentheses becomes $(\dots)(2y \dots)^2 - (\dots)(2y \dots)^2$, which is divisible by $4y^2$. Dividing by the external $2y$ leaves a factor of $2y$.

We now examine the case for when m is odd, letting $m = 2n + 1$:

$$\psi_{2n+1} = \psi_{n+2}\psi_n^3 - \psi_{n-1}\psi_{n+1}^3.$$

- **Case 2.1: m odd, n even.** By the inductive assumption, $\psi_n, \psi_{n+2} \in 2y\mathbb{Z}[x]$. Thus $\psi_{n+2}\psi_n^3 = 16y^4 f_3$. Substituting $y^2 = x^3 + Ax + B$ ensures the term is in $\mathbb{Z}[x]$. The same applies to the second term since $n \pm 1$ are odd.
- **Case 2.2: m odd, n odd.** By symmetry, the parities of the terms swap, but the result remains strictly in $\mathbb{Z}[x]$ after substituting for the y factors.

Thus in any case the proposition is fulfilled □

This proposition is useful, in particular when looking at (odd) prime torsion points, we need only look at roots of a univariate polynomial. A simple corollary describing the auxiliary polynomials defined by:

$$\begin{aligned}\phi_n &= x\psi_n^2 - \psi_{n+1}\psi_{n-1}, \\ 4y\omega_n &= \psi_{n+2}\psi_{n-1}^2 - \psi_{n-2}\psi_{n+1}^2.\end{aligned}$$

Corollary 2.3. *For every natural number n :*

i $\phi_n \in \mathbb{Z}[x]$.

ii $\omega_n \in y \cdot \mathbb{Z}[x]$ for n odd, and $\omega_n \in \mathbb{Z}[x]$ for n even.

Proof. We prove cases *i* and *ii* separately.

Proof of i. By definition $\phi_n = x\psi_n^2 - \psi_{n-1}\psi_{n+1}$. If n is odd, $\psi_n \in \mathbb{Z}[x]$ and $\psi_{n\pm 1} \in 2y\mathbb{Z}[x]$, making $\psi_{n-1}\psi_{n+1} \in 4y^2\mathbb{Z}[x]$. If n is even, $\psi_n \in 2y\mathbb{Z}[x]$ and $\psi_{n\pm 1} \in \mathbb{Z}[x]$. In both cases, y^2 substitutions leave $\phi_n \in \mathbb{Z}[x]$. $\square$

We next prove the case for ω_n .

Proof of ii. Using the recurrence $\omega_n = \frac{\psi_{n+2}\psi_{n-1}^2 - \psi_{n-2}\psi_{n+1}^2}{4y}$:

- **Subcase 2.1: n odd.** The numerator is of the form $4y^2(f)$ where

$$f = \psi_{n+2}\psi_{n-1}^2 - \psi_{n-2}\psi_{n+1}^2.$$

We claim f is a function in $y \cdot \mathbb{Z}[x]$. By Proposition 2.2, since n is even, f is of the form $(f_1)(2y \cdot f_2)^2 - (f_3)(2y \cdot f_4)^2 = 4y^2(f_1)(f_2)^2 - 4y^2(f_3)(f_4)^2 = 4y^2(f_1f_2^2 - f_3f_4^2)$. For $f_i \in \mathbb{Z}[x]$. Dividing by $4y$ leaves $y \cdot (f_1f_2^2 - f_3f_4^2) \in y \cdot \mathbb{Z}[x]$.

- **Subcase 2.2: n even.** The numerator is of the form $2y(f_{n+2}\psi_{n-1}^2 - f_{n-2}\psi_{n+1}^2)$ where $f_n = \frac{\psi}{2y}$. Dividing by $4y$ yields

$$\frac{1}{2}(f_{n+2}\psi_{n-1}^2 - f_{n-2}\psi_{n+1}^2). \quad (\dagger)$$

To show this is in $\mathbb{Z}[x]$, we must show the $f_{n+2}\psi_{n-1}^2 - f_{n-2}\psi_{n+1}^2$ is divisible by 2. By Lemma 2.4 which we have yet to prove, the leading coefficients for f_{n+2} and f_{n-2} is given by:

$$\frac{n+2}{2}(x^2+a)^{\cdots} = \left(\frac{n}{2}+1\right)(x^2+a)^{\cdots} \quad \text{and} \quad \frac{n-2}{2}(x^2+a)^{\cdots} = \left(\frac{n}{2}-1\right)(x^2+a)^{\cdots}$$

Since $(\frac{n}{2}+1) \equiv (\frac{n}{2}-1) \pmod{2}$, we let $c \in \{0, 1\}$ denote this common coefficient. Applying the congruences from the Lemma, we have:

$$\psi_{n\pm 2} \equiv c(x^2+a)^{\frac{(n\pm 2)^2-4}{4}} \pmod{2}$$

$$\psi_{n\pm 1}^2 \equiv \left((x^2+a)^{\frac{(n\pm 1)^2-1}{4}}\right)^2 = (x^2+a)^{\frac{(n\pm 1)^2-1}{2}} \pmod{2}$$

Substituting these into the expression in 2.3 yields:

$$(2.3) \equiv c(x^2 + a)^{\frac{n^2+4n}{4}}(x^2 + a)^{\frac{n^2-2n}{2}} - c(x^2 + a)^{\frac{n^2-4n}{4}}(x^2 + a)^{\frac{n^2+2n}{2}} \pmod{2}$$

By summing the exponents for the first term:

$$\frac{n^2 + 4n + 2(n^2 - 2n)}{4} = \frac{3n^2}{4}$$

And for the second term:

$$\frac{n^2 - 4n + 2(n^2 + 2n)}{4} = \frac{3n^2}{4}$$

Since the coefficients and the exponents are identical modulo 2, the terms cancel:

$$(2.3) \equiv c(x^2 + a)^{\frac{3n^2}{4}} - c(x^2 + a)^{\frac{3n^2}{4}} \equiv 0 \pmod{2}$$

This confirms that 2.3 is divisible by 2, which cancels the remaining factor of 2 in the denominator of the recurrence for ω_n , ensuring $\omega_n \in \mathbb{Z}[x]$. □

Thus all conditions have been satisfied. □

We now must prove for even n , $f_{n+2}\psi_{n-1}^2 - f_{n-2}\psi_{n+1}^2$ is congruent to 0 (mod 2).

Lemma 2.4. *For the division polynomials ψ_n associated with $y^2 = x^3 + ax + b$, the following congruences hold modulo 2:*

1. *If n is odd, $\psi_n \equiv (x^2 + a)^{(n^2-1)/4} \pmod{2}$.*
2. *If n is even, $f_n = \frac{\psi_n}{2y} \equiv \frac{n}{2}(x^2 + a)^{(n^2-4)/4} \pmod{2}$.*

Proof. We proceed by induction on n . The base cases for small n are easily verified. Assume the lemma holds for all values up to $n - 1$.

Case 1: n is odd. Let $n = 2m + 1$. We use the recurrence:

$$\psi_{2m+1} = \psi_{m+2}\psi_m^3 - \psi_{m-1}\psi_{m+1}^3$$

- **Subcase 1.1: m is even.** Then m and $m + 2$ are even. By Proposition 2.2 on page 8, ψ_m and ψ_{m+2} are both multiples of $2y$. Thus, $\psi_{m+2}\psi_m^3 \equiv 0 \pmod{2}$. The expression reduces to:

$$\psi_{2m+1} \equiv \psi_{m-1}\psi_{m+1}^3 \pmod{2}$$

Since $m \pm 1$ are odd, we apply the inductive assumption:

$$\psi_{2m+1} \equiv (x^2 + a)^{\frac{(m-1)^2-1}{4}} \left((x^2 + a)^{\frac{(m+1)^2-1}{4}} \right)^3 = (x^2 + a)^{\frac{4m^2+4m}{4}} = (x^2 + a)^{\frac{m^2-1}{4}}.$$

- **Subcase 1.2: m is odd.** Then $m - 1$ and $m + 1$ are even, making the second term vanish modulo 2. A symmetric calculation using $\psi_{m+2}\psi_m^3$ yields the same result (since $-1 \equiv 1 \pmod{2}$).

Case 2: n is even. Let $n = 2m$. Let $f_{2m} = 2y\psi_{2m}$. We use the recurrence:

$$f_{2m} = \psi_m(f_{m-1}^2\psi_{m+2} - f_{m+1}^2\psi_{m-2})$$

- **Subcase 2.1: m is even.** By inductive assumption, $\psi_m \equiv 0 \pmod{2}$ (since it contains a factor $2y$). Thus $f_{2m} \equiv 0 \pmod{2}$, and $\frac{n}{2} = m \equiv 0 \pmod{2}$. Matching the coefficient $\frac{n}{2}$.
- **Subcase 2.2: m is odd.** Then $\frac{n}{2} = m \equiv 1 \pmod{2}$. Here ψ_m is odd, and exactly one of $\frac{m-1}{2}$ or $\frac{m+1}{2}$ is odd. If $\frac{m+1}{2}$ is odd, then $f_{m+1} \equiv (x^2 + a)^\cdots$ and $f_{m-1} \equiv 0 \pmod{2}$. The recurrence then simplifies to:

$$f_{2m} \equiv \psi_m f_{m+1}^2 \psi_{m-2} \pmod{2}$$

Summing the exponents from the inductive assumption:

$$\frac{m^2 - 1}{4} + 2\frac{(m+1)^2 - 4}{4} + \frac{(m-2)^2 - 4}{4} = \frac{4m^2 - 4}{4} = m^2 - 1.$$

Since $m^2 - 1 = \frac{(2m)^2 - 4}{4}$, the congruence holds.

□

2.2 Multiplication-by- n Map

The results of the previous section give us insight on how nicely the division polynomials work. In this section we will discuss why these are important and are being studied. The following lemma shows us the degree of the leading x term, again this will be useful in the study of torsion points.

Lemma 2.5 (Degrees and Leading Terms, cf. [Was08] Lemma 3.5). *For the division polynomials ψ_m and ϕ_m :*

1. $\psi_m^2(x)$ has degree $m^2 - 1$ with leading coefficient m^2 .
2. $\phi_m(x)$ has degree m^2 with leading coefficient 1.

Proof. We proceed by induction on m . The base cases $m = 1, 2, 3, 4$ satisfy the lemma by inspection. Assume the lemma holds for all integers $k < m$. We treat the leading terms of ψ_m as having “degree” $(m^2 - 1)/2$ by weighting the y term to be $3/2$ since $y^2 = x^3 + Ax + B$.

We first examine the leading term of ψ_m^2 . We examine the case where $m = 2n + 1$. The recurrence relation is:

$$\psi_{2n+1} = \psi_{n+2}\psi_n^3 - \psi_{n-1}\psi_{n+1}^3$$

Assume n is even. By the inductive assumption, the leading terms (lt) are:

$$\begin{aligned} \text{lt}(\psi_{n+2}) &= \frac{n+2}{2}(2y)x^{\frac{(n+2)^2-4}{2}}, & \text{lt}(\psi_n) &= \frac{n}{2}(2y)x^{\frac{n^2-4}{2}} \\ \text{lt}(\psi_{n+1}) &= (n+1)x^{\frac{(n+1)^2-1}{2}}, & \text{lt}(\psi_{n-1}) &= (n-1)x^{\frac{(n-1)^2-1}{2}} \end{aligned}$$

Substituting these into the recurrence and using $y^2 \sim x^3$:

- The first term $\psi_{n+2}\psi_n^3$ has leading coefficient $(n+2)n^3$ and degree $\frac{4n^2+4n}{2}$.
- The second term $\psi_{n-1}\psi_{n+1}^3$ has leading coefficient $(n-1)(n+1)^3$ and degree $\frac{4n^2+4n}{2}$.

Subtracting the coefficients:

$$(n+2)n^3 - (n-1)(n+1)^3 = n^4 + 2n^3 - (n^4 + 2n^3 - 2n - 1) = 2n + 1$$

Thus, $\text{lt}(\psi_{2n+1}) = (2n+1)x^{\frac{(2n+1)^2-1}{2}}$. Squaring this yields a degree of $m^2 - 1$ and leading coefficient m^2 .

We now examine the case where $m = 2n$. The recurrence relation is:

$$\psi_{2n} = \frac{\psi_n}{2y} (\psi_{n+2}\psi_{n-1}^2 - \psi_{n-2}\psi_{n+1}^2)$$

Assume n is even. By the inductive assumption, we utilize:

$$\begin{aligned} \text{lt}(\psi_n) &= \frac{n}{2}(2y)x^{\frac{n^2-4}{4}} \\ \text{lt}(\psi_{n+2}) &= \frac{n+2}{2}(2y)x^{\frac{(n+2)^2-4}{4}}, & \text{lt}(\psi_{n-1}) &= x^{\frac{(n-1)^2-1}{4}}, \\ \text{lt}(\psi_{n-2}) &= \frac{n-2}{2}(2y)x^{\frac{(n-2)^2-4}{4}}, & \text{lt}(\psi_{n+1}) &= x^{\frac{(n+1)^2-1}{4}}. \end{aligned}$$

Substituting these into the parentheses (with $2y \sim 2x^{3/2}$):

- **First term:** $\text{lt}(\psi_{n+2}\psi_{n-1}^2) = \frac{n+2}{2}(2x^{3/2})x^{\frac{n^2+4n}{4}} \cdot x^{\frac{n^2-2n}{2}} = (n+2)x^{\frac{3n^2+6}{4}}$.
- **Second term:** $\text{lt}(\psi_{n-2}\psi_{n+1}^2) = \frac{n-2}{2}(2x^{3/2})x^{\frac{n^2-4n}{4}} \cdot x^{\frac{n^2+2n}{2}} = (n-2)x^{\frac{3n^2+6}{4}}$.

The leading term of the expression in parentheses is the difference:

$$(n+2)x^{\frac{3n^2+6}{4}} - (n-2)x^{\frac{3n^2+6}{4}} = 4x^{\frac{3n^2+6}{4}}$$

Multiplying by the external factor $\frac{\psi_n}{2y} \sim \frac{n}{2}x^{\frac{n^2-4}{4}}$:

$$\text{lt}(\psi_{2n}) = \left(\frac{n}{2}x^{\frac{n^2-4}{4}}\right) \left(4x^{\frac{3n^2+6}{4}}\right) = 2nx^{\frac{4n^2+2}{4}}$$

Solving for the reduced polynomial f_m where $m = 2n$:

$$\text{lt}(f_m) = \frac{\text{lt}(\psi_m)}{2y} = \frac{2nx^{\frac{2n^2+1}{2}}}{2x^{3/2}} = nx^{n^2-1} = \frac{m}{2}x^{\frac{m^2-4}{4}}$$

Squaring the full division polynomial $\psi_m = 2yf_m$ finally gives:

$$\text{lt}(\psi_m^2) = (4x^3) \left(\frac{m^2}{4}x^{\frac{m^2-4}{2}}\right) = m^2x^{\frac{m^2+2}{2}}$$

This confirms the leading coefficient is m^2 , matching the inductive step for all m .

We next prove the case of ϕ_m . By definition, $\phi_m = x\psi_m^2 - \psi_{m-1}\psi_{m+1}$. Using the results from Step 1:

- $\text{lt}(x\psi_m^2) = x \cdot (m^2x^{m^2-1}) = m^2x^{m^2}$.
- $\text{lt}(\psi_{m-1}\psi_{m+1}) = (m-1)x^{\frac{(m-1)^2-1}{2}} \cdot (m+1)x^{\frac{(m+1)^2-1}{2}}$.

The product of the coefficients is $(m-1)(m+1) = m^2 - 1$. The sum of the degrees is:

$$\frac{m^2 - 2m + 1 - 1 + m^2 + 2m + 1 - 1}{2} = \frac{2m^2}{2} = m^2$$

Combining these results:

$$\text{lt}(\phi_m) = m^2x^{m^2} - (m^2 - 1)x^{m^2} = (m^2 - m^2 + 1)x^{m^2} = x^{m^2}$$

Thus, ϕ_m is monic with degree m^2 . □

We can now discuss the main theorem and motivation behind division polynomials, that is we can define a general multiplication-by- n map for points on a curve.

Theorem 2.6. *Let $P = (x, y) \in C$ such that $nP \neq \mathcal{O}$. Then*

$$[n]P = \left(\frac{\phi_n(P)}{\psi_n(P)^2}, \frac{\omega_n(P)}{\psi_n(P)^3}\right).$$

Proof. We again use the powerful tool of induction on n . For $n = 1$, the statement is trivial since $\psi_1 = 1, \phi_1 = x, \omega_1 = y$. For $n = 2$, the formula follows immediately from the standard duplication formula on the Weierstrass curve. Recall from Section 1.3.1:

$$x([2]P) = \frac{x^4 - 2bx^2 - 8cx + b^2}{4x^3 + 4ax + 4b}.$$

Calculating ϕ_2/ψ_2^2 using the definitions $\phi_2 = x\psi_2^2 - \psi_3\psi_1$ and $\psi_2 = 2y$ yields:

$$x([2]P) = \frac{x^4 - 2bx^2 - 8cx + b^2}{4y^2}.$$

Notice the denominators are different but by substituting $y^2 = x^3 + ax + b$ in the denominator, we find they are equivalent.

For the inductive step, we utilize the addition formulas. Specifically, we assume the formula holds for n and $n - 1$. From the group law, we have the relation:

$$x_{n+1} - x_{n-1} = -\frac{\psi_{n-1}\psi_{n+1}}{\psi_n^2}(x_n - x)^{-2},$$

where x_k denotes the x -coordinate of $[k]P$. By the definition of ϕ_n , we have $x_n = \phi_n/\psi_n^2$. Substituting this into the left-hand side and manipulating the recurrence relations (from (2.1) and (2.2)) allows us to verify that x_{n+1} satisfies the required form ϕ_{n+1}/ψ_{n+1}^2 . The y -coordinate follows similarly from the fact that we are working from a curve $y^2 = x^3 + ax + b$ $\square$

Corollary 2.7. *Let $n \in \mathbb{Z}^+$. A point $P = (x, y) \in C$ is an n -torsion point if and only if $\psi_n(P) = 0$. That is, the roots of the division polynomial ψ_n are exactly the x -coordinates of the points in $E[n] \setminus \{\mathcal{O}\}$.*

Proof. By definition, a point P is an n -torsion point if and only if $[n]P = \mathcal{O}$. In the context of the rational functions defined in Theorem 2.6, the point at infinity $\mathcal{O}$ corresponds to the pole of the coordinate functions.

The x -coordinate of the multiplied point is given by the rational function:

$$x([n]P) = \frac{\phi_n(P)}{\psi_n(P)^2}$$

Note that ϕ_n and ψ_n^2 have distinct roots. Consequently, the x -coordinate becomes infinite if and only if the denominator vanishes. Specifically:

$$[n]P = \mathcal{O} \iff \psi_n(P)^2 = 0 \iff \psi_n(P) = 0.$$

Thus, the non-trivial n -torsion points are precisely those points whose x -coordinates are roots of the n -th division polynomial ψ_n . $\square$

2.3 Structure of the Torsion Subgroup

The degree of ψ_n^2 tells us about the number of torsion points; specifically, since the map $[n]$ is given by quotients of these polynomials, the kernel of this map the group of n -torsion points, corresponds

to the poles of the coordinates of $[n]P$, or equivalently, the zeros of the denominator ψ_n^2 . We will also discuss a special map that defines an isomorphism between our group and another. In this section we will prove the following theorem.

Theorem 2.8 (Structure of the Torsion Subgroup over $\mathbb{C}$). *Let E be an elliptic curve over $\mathbb{C}$ and let n be a positive integer. The subgroup of n -torsion points $E[n]$ satisfies:*

$$E[n] \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.$$

To understand the group structure of E , we first examine the analytic properties of the Weierstrass $\wp$ -function. Given a lattice $\Lambda \subset \mathbb{C}$ generated by a basis $\{\omega_1, \omega_2\}$, the $\wp$ -function is defined as follows:

Definition 2.9 (Weierstrass $\wp$ -function).

$$\wp(z; \Lambda) = \frac{1}{z^2} + \sum_{\omega \in \Lambda \setminus \{0\}} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right)$$

The function $\wp(z)$ is a meromorphic, doubly periodic function with respect to Λ , meaning $\wp(z + \omega) = \wp(z)$ for all $\omega \in \Lambda$. It has a pole of order 2 at each lattice point and is even (i.e. $\wp(z) = \wp(-z)$). This periodicity implies that $\wp$ is essentially a function defined on the quotient space $\mathbb{C}/\Lambda$. These properties are proven in Appendix B on page 34

The connection between this analytic construction and the algebraic geometry of elliptic curves is given by the map Ξ :

Proposition 2.10. *The map $\Xi : \mathbb{C}/\Lambda \rightarrow E(\mathbb{C})$ defined by*

$$\Xi(z) = \begin{cases} (\wp(z), \wp'(z)) & \text{if } z \notin \Lambda \\ \mathcal{O} & \text{if } z \in \Lambda \end{cases}$$

is an isomorphism of abelian groups.

Proof. To establish that Ξ is a group isomorphism, we verify three key properties: well-definedness, the homomorphism property, and bijectivity.

1. Well-definedness: The Weierstrass $\wp$ -function and its derivative satisfy the algebraic relation:

$$(\wp'(z))^2 = 4\wp(z)^3 - a\wp(z) - b$$

for all $z \in \mathbb{C} \setminus \Lambda$. By definition, the image of any point $z \notin \Lambda$ under Ξ satisfies the affine equation of the elliptic curve E . Since $\Xi(\Lambda) = \mathcal{O}$ (the point at infinity), the map is well-defined from the quotient $\mathbb{C}/\Lambda$ to $E(\mathbb{C})$.

2. Group Homomorphism: We must show that $\Xi(z_1 + z_2) = \Xi(z_1) + \Xi(z_2)$, where the right side follows the chord-and-tangent group law on E . Recall that $P_1 + P_2 + P_3 = \mathcal{O}$ on an elliptic curve if and only if the points are collinear.

Consider the elliptic function $f(z) = \wp'(z) - m\wp(z) - c$, where $y = mx + c$ is the line passing through $\Xi(z_1)$ and $\Xi(z_2)$. The function $f(z)$ has a triple pole at $z = 0$ and three zeros at z_1, z_2 , and a third point z_3 . According to the theory of elliptic functions, the sum of the zeros of a meromorphic function on a torus must be congruent to the sum of its poles modulo the lattice. Thus:

$$z_1 + z_2 + z_3 \equiv 0 \pmod{\Lambda}$$

This geometric alignment on the torus corresponds exactly to the collinearity condition on the curve, ensuring that Ξ preserves the group operation.

3. Bijectivity: For any $x \in \mathbb{C}$, the function $g(z) = \wp(z) - x$ is an elliptic function of order 2, meaning it takes the value x exactly twice in the fundamental domain (at some z_0 and $-z_0$). Since $\wp'(z)$ is an odd function, $\wp'(z_0)$ and $\wp'(-z_0)$ provide the two possible y -roots for the cubic $y^2 = 4x^3 - ax - b$. This ensures that every point (x, y) on the curve has a unique preimage $z \in \mathbb{C}/\Lambda$. Combined with $\Xi(0) = \mathcal{O}$, we conclude that Ξ is a bijection. $\square$

Under the isomorphism Ξ , the n -torsion subgroup $E[n]$ corresponds exactly to the n -torsion subgroup of the torus, $(\mathbb{C}/\Lambda)[n]$. These are points on the fundamental parallelogram shown in

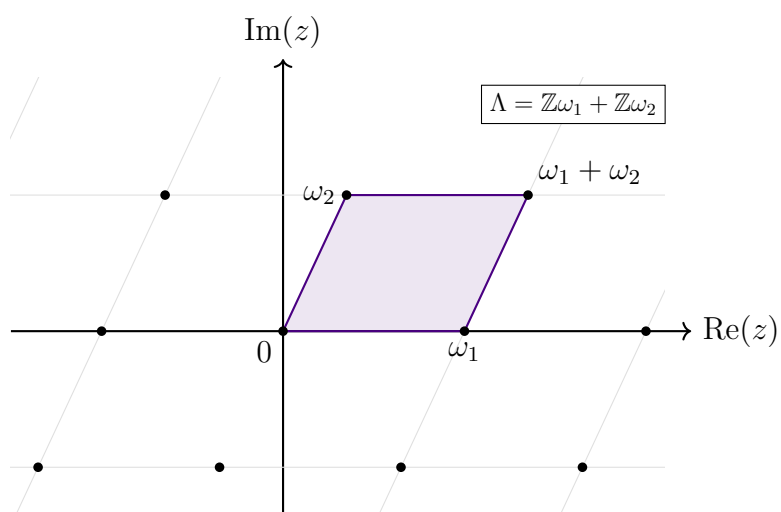
Figure 2.1 A point $z + \Lambda$ is an n -torsion point if and only if $nz \in \Lambda$. This requires:

$$nz = a\omega_1 + b\omega_2 \implies z = \frac{a}{n}\omega_1 + \frac{b}{n}\omega_2$$

for some integers a, b . Distinguishing these points modulo Λ , we obtain the set:

$$(\mathbb{C}/\Lambda)[n] = \left\{ \frac{a}{n}\omega_1 + \frac{b}{n}\omega_2 + \Lambda : 0 \leq a, b < n \right\}.$$

This set is generated by the two independent elements $\frac{1}{n}\omega_1$ and $\frac{1}{n}\omega_2$, both of order n . Consequently, the group structure is $(\mathbb{C}/\Lambda)[n] \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$. Since Ξ is a group isomorphism, it follows immediately that $E[n]$ shares this structure. Thus Theorem 2.8 on the preceding page is proven.

Figure 2.1: Torus $\mathbb{C}/\Lambda$ with fundamental parallelogram.

Chapter 3

Complex Multiplication and Special Maps

In the previous chapter, we derived the multiplication maps for integers $[n]$. However, our specific curve $y^2 = x^3 + x$ admits a larger ring of endomorphisms isomorphic to the Gaussian integers $\mathbb{Z}[i]$.

In this chapter, we derive the explicit maps for these complex endomorphisms.

3.1 Derivation of the Gaussian Integer Multiplication Map

The curve $C : y^2 = x^3 + x$ possesses an automorphism defined by the imaginary unit i :

$$[i] : (x, y) \mapsto (-x, iy).$$

Consequently, there exist Gaussian torsion points such that for $\pi \in \mathbb{Z}[i]$, $\pi P = \mathcal{O}$.

Example 3.1. Consider $\pi = 1 + i$. A point P is in the kernel if $(1 + i)P = P + [i]P = \mathcal{O}$. The norm is $N(1 + i) = 2$, so these points lie inside $E[2]$.

3.1.1 Derivation of ψ_π and ϕ_π

Let $\pi = n + im$ be a Gaussian integer. The point πP is given by $\pi P = nP + [i](mP)$. Let

$P_1 = nP$ and $P_2 = [i](mP)$. Substituting the standard division polynomials:

$$P_1 = \left(\frac{\phi_n}{\psi_n^2}, \frac{\omega_n}{\psi_n^3} \right) \tag{3.1}$$

$$P_2 = [i] \left(\frac{\phi_m}{\psi_m^2}, \frac{\omega_m}{\psi_m^3} \right) = \left(-\frac{\phi_m}{\psi_m^2}, \frac{i\omega_m}{\psi_m^3} \right) \tag{3.2}$$

Using the addition law for $y^2 = x^3 + x$, the x -coordinate of the sum $P_3 = P_1 + P_2$ is:

$$x_3 = \frac{(x_1 + x_2)(1 + x_1x_2) - 2y_1y_2}{(x_2 - x_1)^2}.$$

The Denominator and Numerator

We now must substitute our x, y points from the n, m derivations, to find $(n + mi)P = \pi P$. For simplicity sake, we will split this into numerator N and denominator D .

D : for this case we will look specifically at $(x_2 - x_1)^2$. Substituting from (3.1) and (3.2), we see:

$$D = (x_2 - x_1)^2 = \left(-\frac{\phi_m}{\psi_m^2} - \frac{\phi_n}{\psi_n^2} \right)^2 = \left(-\frac{\phi_m\psi_n^2 + \phi_n\psi_m^2}{\psi_m^2\psi_n^2} \right)^2 = \frac{(\phi_m\psi_n^2 + \phi_n\psi_m^2)^2}{\psi_m^4\psi_n^4}$$

N : for this case we will look at $(x_1 + x_2)(1 + x_1x_2) - 2y_1y_2$. Substituting, we see:

$$(x_1 + x_2)(1 + x_1x_2) - 2y_1y_2 = \left(\frac{\phi_n}{\psi_n^2} - \frac{\phi_m}{\psi_m^2} \right) \left(1 + \frac{\phi_n}{\psi_n^2} \cdot -\frac{\phi_m}{\psi_m^2} \right) - 2i\frac{\omega_n}{\psi_n^3} \cdot \frac{\omega_m}{\psi_m^3}.$$

Breaking this down even more we will look at the $(x_1 + x_2)$ term which simplifies as such:

$$\left(\frac{\phi_n}{\psi_n^2} - \frac{\phi_m}{\psi_m^2} \right) = \left(\frac{\phi_n\psi_m^2 - \phi_m\psi_n^2}{\psi_m^2\psi_n^2} \right) \quad (3.3)$$

Looking at $(1 + x_1x_2)$:

$$\left(1 + \frac{\phi_n}{\psi_n^2} \cdot -\frac{\phi_m}{\psi_m^2} \right) = \left(\frac{\psi_n^2\psi_m^2 - \phi_m\phi_n}{\psi_m^2\psi_n^2} \right) \quad (3.4)$$

We multiply (3.3) and (3.4):

$$\left(\frac{\phi_n\psi_m^2 - \phi_m\psi_n^2}{\psi_m^2\psi_n^2} \right) \cdot \left(\frac{\psi_n^2\psi_m^2 - \phi_m\phi_n}{\psi_m^2\psi_n^2} \right) = \frac{\phi_n\psi_n^2\psi_m^4 - \phi_m\psi_n^4\psi_m^2 - \phi_n^2\phi_m\psi_m^2 + \phi_m^2\phi_n\psi_n^2}{\psi_m^4\psi_n^4}$$

Finally we look at $2y_1y_2$:

$$2 \left(\frac{\omega_n}{\psi_n^3} \right) \left(\frac{i\omega_m}{\psi_m^3} \right) = \frac{2i\omega_n\omega_m\psi_m\psi_n}{\psi_n^4\psi_m^4}$$

Adding this together we get

$$N = \frac{\phi_n\psi_n^2\psi_m^4 - \phi_m\psi_n^4\psi_m^2 - \phi_n^2\phi_m\psi_m^2 + \phi_n\phi_m^2\psi_n^2 - 2i\omega_n\omega_m\psi_n\psi_m}{\psi_m^4\psi_n^4}.$$

Notice that N and D have common denominators, put it all together to find $\frac{N}{D}$:

$$x(\pi P) = \frac{\phi_n\psi_n^2\psi_m^4 - \phi_m\psi_n^4\psi_m^2 - \phi_n^2\phi_m\psi_m^2 + \phi_n\phi_m^2\psi_n^2 - 2i\omega_n\omega_m\psi_n\psi_m}{(\phi_m\psi_n^2 + \phi_n\psi_m^2)^2}$$

While in the denominator $\sqrt{D} =: \Psi_\pi$ derived from the addition law correctly vanishes on the π -torsion points, its degree $N(\pi) - 1$ indicates that it is the product of the π -division polynomial and its conjugate, $\psi_\pi \cdot \psi_{\bar{\pi}}$. This is expected, as the x-coordinate addition law is symmetric with respect to π and $\bar{\pi}$. To apply the Eisenstein criterion, we must isolate the irreducible factor $\psi_\pi \in \mathbb{Z}[i][x]$ whose leading coefficient is π . As seen in the following examples, this factor possesses a constant term that is a unit in $\mathbb{Z}[i]$, while all intermediate coefficients are divisible by π .

For completion we define:

$$\Psi_\pi := \phi_m \psi_n^2 + \phi_n \psi_m^2 \quad (3.5)$$

$$\Phi_\pi := \phi_n \psi_n^2 \psi_m^4 - \phi_m \psi_n^4 \psi_m^2 - \phi_n^2 \phi_m \psi_m^2 + \phi_n \phi_m^2 \psi_n^2 - 2i\omega_n \omega_m \psi_n \psi_m. \quad (3.6)$$

We derive Ω_π from our x value, substituting into our curve equation.

$$\begin{aligned} \Omega_\pi &= \omega_n \psi_m [\psi_m^2 (\phi_n \psi_m^2 + \phi_m \psi_n^2)^2 - (\phi_n \psi_m^2 - \phi_m \psi_n^2)^2] \\ &\quad - i\omega_m \psi_n [\psi_n^2 (\phi_n \psi_m^2 + \phi_m \psi_n^2)^2 - (\phi_n \psi_m^2 - \phi_m \psi_n^2)^2] \\ &\quad + 4i\omega_n \omega_m (\phi_n \psi_m^2 + \phi_m \psi_n^2). \end{aligned} \quad (3.7)$$

So we have derived a map:

$$[\pi]P = \left(\frac{\Phi_\pi}{\Psi_\pi^2}, \frac{\Omega_\pi}{\Psi_\pi^3} \right).$$

The following lemma will hold for the curve $y^2 = x^3 + x$.

Lemma 3.2. For odd n , $\psi_n = nx^{(n^2-1)/2} \dots + 1_{\mathbb{Z}[i]}$ where $1_{\mathbb{Z}[i]} \in \mathbb{Z}[i]^\times$.

Proof. We have proven in Lemma 2.5 that the leading coefficient is as shown. We now must prove that the trailing coefficient is $1_{\mathbb{Z}[i]}$. We Prove this by induction, from the definition of the curve $\psi_1 = 1$ and for the curve $y^2 = x^3 + x$, $\psi_3 = 3x^4 + 6x^2 - 1$. Assume this condition holds for $2n - 1$ iterations. This breaks into two cases.

1. **Case 1: n is even** If n is even then looking at $\psi_{n_2} \psi_n^3 - \psi_{n-1} \psi_{n+1}^3$, from the inductive assumption, ψ_{n-1} has a trailing coefficient $1_{\mathbb{Z}[i]}$ and so does ψ_{n+1}^3 so multiplying keeps this trailing coefficient. As we have proven back in Proposition 2.2 on page 8, ψ_{2m} has a factor of y , indeed, the even part $\psi_{n_2} \psi_n^3$ does not have a constant term. Since $\psi_{n-1} \psi_{n+1}^3$ has a constant term $1_{\mathbb{Z}[i]}$ and $\psi_{n_2} \psi_n^3$ has no constant term, $\psi_{n_2} \psi_n^3 - \psi_{n-1} \psi_{n+1}^3$ has a constant term $1_{\mathbb{Z}[i]}$
2. **Case 2: n is odd** Notice by symmetry, the odd case can be turned into the even case by multiplying by -1 . Thus the same holds

□

3.1.2 Arithmetic Properties and the Eisenstein Property

We now examine the reduction of Ψ_π . This serves as the foundation for proving that the torsion fields generate totally ramified extensions. Using a computer for prime π , we compute¹:

$$\Psi_{1+2i} = 5x^4 + 2x^2 + 1,$$

$$\Psi_{2+3i} = 13x^{12} - 26x^{10} + 39x^8 + 228x^6 + 235x^4 + 22x^2 + 1.$$

Ψ_{1+2i} can be factored into $((1 + 2i)x^2 + 1)((1 - 2i)x^2 + 1)$. These factors clearly are irreducible by Eisenstein criterion with prime $\pi = 1 + 2i$ and $\bar{\pi}$ respectively. Ψ_{2+3i} can be factored as such:

$$\begin{aligned} \Psi_{2+3i} = & ((2 + 3i)x^6 + (4 - 7i)x^4 + (10 - 11i)x^2 - i) \cdot \\ & \cdot ((2 - 3i)x^6 + (4 + 7i)x^4 + (10 + 11i)x^2 + i). \end{aligned}$$

We will show that this satisfies Eisenstein criterion. We check individually that $(4 - 7i) \mid (2 + 3i)$, and $(10 - 11i) \mid (2 + 3i)$. Calculating $(2 + 3i)(-1 - 2i) = (4 - 7i)$ this implies that $(4 - 7i) \mid (2 + 3i)$, calculate $(2 + 3i)(-1 - 4i) = (10 + 11i)$. Thus $(2 + 3i)$ (which is a Gaussian prime) divides all but the constant coefficient and $\pi^2 \nmid \pi$, so it satisfies Eisenstein. The same calculation can be done on the second factor, we claim that all coefficients are complex conjugates of the first factor.

Thus we claim that $\Psi_\pi = \psi_\pi \psi_{\bar{\pi}}$ where ψ_π is irreducible for Gaussian prime π . In order to prove this, we must first show that Ψ can be split into these polynomials and that these two factors are irreducible.

A point P is a torsion point of Ψ_π if and only if $x([m]P + [ni]P)$ vanishes. Thus occurs if and only if $[m]P + [ni]P = \mathcal{O}$

We would like to study how this torsion group looks and how we can interact with it. The following theorem builds from Theorem 2.8 on page 15.

Theorem 3.3 (Structure of the Gaussian Torsion Subgroup). *For p -torsion points on the curve $y^2 = x^3 + x$:*

(1) *If $p \equiv 1 \pmod{4}$, the p -torsion subgroup is*

$$E[p] \cong E[\pi] \oplus E[\bar{\pi}]$$

for Gaussian integer π such that $p = \pi\bar{\pi}$.

(2) *If $p \equiv 3 \pmod{4}$, the p -torsion subgroup is indecomposable.*

¹The python script used for calculating can be found in Appendix A on page 28

Proof. We first consider the case (1). Let $V = E[p]$. Consider the multiplication by i map we have been studying in this section. $[i]P = (-x, iy)$, taking this map twice $[i]^2P = (x, -y) = -P$, thus $[i]^2 = [-1]$. Think of the multiplication by i map as a matrix:

$$[i]^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I.$$

Thus $[i]^2 + I = 0$, finding eigenvalues and eigenvectors of this matrix leads to the characteristic polynomial $\varphi^2 + 1 = 0$ for some φ . Since $p \equiv 1 \pmod{2}$, there exists some λ such that $\lambda^2 \equiv -1 \pmod{p}$ (i.e. $\varphi^2 + 1 = (\varphi - \lambda)(\varphi + \lambda)$.) Thus the eigenvalues are $\lambda, -\lambda$ and since they are distinct (since $\lambda \neq 0$), they are linearly independent, and:

$$\begin{aligned} V_\lambda &= \{P \in E[p] \mid [i]P = \lambda P\} \\ V_{-\lambda} &= \{P \in E[p] \mid [i]P = -\lambda P\}. \end{aligned}$$

Thus for V_λ , $[i]P = [\lambda]P$ this implies $[i]P - [\lambda]P = 0$. So $V_\lambda = \ker([i] - [\lambda])$. Similarly for $V_{-\lambda} = \ker([i] + [\lambda])$. The map $[i] + [\lambda]$ corresponds to π , and the map $[i] - [\lambda]$ to $\bar{\pi}$. Thus $V_\lambda = E[\pi]$ and $V_{-\lambda} = E[\bar{\pi}]$. Since $E[\pi]$ and $E[\bar{\pi}]$ are linearly independent eigenspaces:

$$E[p] \cong E[\pi] \oplus E[\bar{\pi}].$$

We now consider the case (2). Let $V = E[p]$. Since p is a prime not dividing the discriminant of the curve, V is a 2-dimensional vector space over the finite field $\mathbb{F}_p$. The endomorphism $[i]$ acts as a linear transformation $\phi : V \rightarrow V$ satisfying the relation $\phi^2 + I = 0$.

The minimal polynomial of ϕ is $P(\varphi) = \varphi^2 + 1$. For V to be decomposable as a $\mathbb{Z}[i]$ -module, it must possess a 1-dimensional invariant subspace (a line) $W \subset V$ such that $\phi(W) \subseteq W$. The existence of such a subspace is equivalent to ϕ having an eigenvalue $\lambda \in \mathbb{F}_p$.

However, the eigenvalues of ϕ are the roots of $\varphi^2 \equiv -1 \pmod{p}$. However, -1 is a quadratic non-residue for all primes $p \equiv 3 \pmod{4}$. Consequently, $\varphi^2 + 1$ is irreducible over $\mathbb{F}_p$, and ϕ has no eigenvectors in V . Since there are no proper non-trivial invariant subspaces, V is indecomposable. In fact, V is a simple $\mathbb{Z}[i]$ -module in this case. $\square$

This will be required in order to show that Φ is reducible. We finally will prove the main theorem of this chapter.

Theorem 3.4. *For a Gaussian prime $\pi \in \mathbb{Z}[i]$, the polynomial Ψ_π factors as $\Psi_\pi = \psi_\pi \psi_{\bar{\pi}}$ in $\mathbb{Z}[i][x]$. Furthermore, the factor ψ_π is irreducible over $\mathbb{Z}[i]$.*

Proof. By Theorem 3.3, the p -torsion subgroup for $p = \pi\bar{\pi}$ decomposes into the direct sum of the kernels of the maps $[\pi]$ and $[\bar{\pi}]$:

$$E[p] \cong E[\pi] \oplus E[\bar{\pi}].$$

The roots of the division polynomial Ψ_π are precisely the x -coordinates of the non-trivial points in $E[p]$. Since the x -coordinate map is symmetric with respect to the eigenspaces of $[i]$ and $[-i]$, the polynomial must partition into factors corresponding to these subspaces. We define $\psi_\pi \in \mathbb{Z}[i][x]$ to be the polynomial whose roots are the x -coordinates of the points in $E[\pi] \setminus \{\mathcal{O}\}$. Because the map $[\pi]$ is defined over $\mathbb{Z}[i]$, the coefficients of ψ_π lie in the Gaussian integers. Thus $\Psi_\pi = \psi_\pi \psi_{\bar{\pi}}$.

We show that ψ_π satisfies the Eisenstein Criterion over the $\mathbb{Z}[i]$ with respect to the prime element π :

1. **Leading Coefficient:** The leading coefficient of the π -division polynomial is π (up to a unit). Since π is prime in $\mathbb{Z}[i]$, it follows that $\pi \mid \pi$ but $\pi^2 \nmid \pi$.
2. **Intermediate Coefficients:** Consider the reduction of the curve E modulo the prime ideal $(\bar{\pi})$. In the finite field $\mathbb{F}_{N(\pi)}$, the endomorphism $[\pi]$ corresponds to the Frobenius map (or a power thereof). Consequently, the only point in the kernel of $[\pi]$ over the algebraic closure is the point at infinity. This implies that $\psi_\pi(x) \equiv \pi x^{\deg \psi} \pmod{\bar{\pi}}$, meaning all coefficients except the leading one are divisible by π .
3. **Constant Term:** By Lemma 3.2, the trailing coefficient of the division polynomials for this curve is a unit in $\mathbb{Z}[i]$ (specifically ± 1 or $\pm i$). Since units are not divisible by any prime in $\mathbb{Z}[i]$, we have $\pi \nmid \psi_\pi(0)$.

Having satisfied these conditions, ψ_π is irreducible over $\mathbb{Z}[i]$. By symmetry of the complex conjugation map, $\psi_{\bar{\pi}}$ is likewise irreducible. □

Chapter 4

Local Analysis and the Eisenstein Criterion

To fully understand the fields generated by torsion points, we must look at the curve "locally"—that is, in the neighborhood of the point at infinity or over local fields like the p -adic numbers.

4.1 Formal Groups

[EDITOR'S NOTE: This section is needed to connect algebra to analysis]

- **Change of Variables:** Introduce $z = -x/y$ and $w = -1/y$. Show how this moves $\mathcal{O}$ to $z = 0$.
- **The Formal Group Law:** Explain that the group law $(z_1, z_2) \mapsto z_3$ can be written as a power series $F(z_1, z_2) = z_1 + z_2 + \dots$ in $\mathbb{Z}[[z_1, z_2]]$.
- **Relevance:** Mention that this allows us to study torsion points by looking at the roots of power series in local fields.

4.2 The Eisenstein Criterion

[EDITOR'S NOTE: Connect division polynomials to field extensions]

- Explain that to find the Galois group of $K(E[p])$, we look at the polynomial ψ_p .
- If ψ_p (or a related polynomial) satisfies the Eisenstein criterion at a prime, the extension is totally ramified.
- This is a standard technique to prove that the field of definition for p -torsion is "large".

Appendix A

Computational Verification

A.1 Python Script for Division Polynomials

a1 The following script calculates the division polynomial Ψ_π .

```
1 import sympy as sp
2 import re
3
4 # --- TERMINAL FORMATTING ---
5 def to_superscript(n):
6     supers = "
7     return "".join(supers[int(d)] for d in str(n))
8
9 def format_poly_clean(poly, var_char='x'):
10     """Restored custom formatter to nicely group coefficients
11         like (1+2i) x """
12     x = sp.symbols(var_char)
13     poly = sp.expand(poly)
14
15     try:
16         poly_obj = sp.Poly(poly, x)
17     except sp.GeneratorsNeeded:
18         s = str(poly).replace("**", "^").replace("*I", "i").
19             replace("I", "i")
20         return re.sub(r"\^(\\d+)", lambda m: to_superscript(m.
21             group(1)), s)
22
23 term_list = poly_obj.terms()
24 if not term_list: return "0"
25
26 terms = []
27 for i, (power_tuple, coeff) in enumerate(term_list):
28     power = power_tuple[0]
29     coeff_str = str(coeff).replace("**", "^").replace("*I", "
```

```

        i").replace("I", "i")
27
28     # Handle sign
29     if coeff_str.startswith("-"):
30         sign = "_-" if i > 0 else "-"
31         val_str = coeff_str[1:]
32     else:
33         sign = "_+" if i > 0 else ""
34         val_str = coeff_str
35
36     # Handle formatting the coefficient block
37     if val_str == "1" and power != 0:
38         final_coeff = ""
39     else:
40         # Wrap in parentheses if it's a complex number or
           expression
41         if ("+" in val_str or "-" in val_str or "/" in
           val_str) and not val_str.startswith("("):
42             final_coeff = f"({val_str})"
43         else:
44             final_coeff = val_str
45
46     # Handle formatting the variable block
47     if power == 0:
48         var_str = ""
49         if final_coeff == "": final_coeff = "1"
50     elif power == 1:
51         var_str = var_char
52     else:
53         var_str = f"{var_char}{to_superscript(power)}"
54
55     terms.append(f"{sign}{final_coeff}{var_str}")
56
57     return "".join(terms)
58
59
60 def is_Z_i_poly(expr):
61     """Verifies if a polynomial's coefficients are strictly
           Gaussian integers."""
62     x = sp.symbols('x')
63     try:
64         coeffs = sp.Poly(expr, x).coeffs()
65     except sp.GeneratorsNeeded:
66         coeffs = [expr]
67

```

```

68     for c in coeffs:
69         try:
70             # Evaluate as a complex number and verify it has no
              decimal/fraction
71             c_val = complex(c)
72             if not (abs(c_val.real - round(c_val.real)) < 1e-9
              and abs(c_val.imag - round(c_val.imag)) < 1e-9):
73                 return False
74         except TypeError:
75             return False
76     return True
77
78 def format_grouped(factors_list):
79     """Takes a list of (factor, exponent) pairs and outputs a
      grouped string."""
80     res = ""
81     for f, p in factors_list:
82         f_str = format_poly_clean(f)
83         if p == 1:
84             res += f"({f_str})"
85         else:
86             res += f"({f_str})^{to_superscript(p)}"
87     return res
88
89 def format_factor_pair(poly, n_map, m_map):
90     """Factors poly over  $\mathbb{Q}(i)$  and systematically restores
      coefficients to  $\mathbb{Z}[i]$ ."""
91     x = sp.symbols('x')
92     pi = n_map + m_map * sp.I
93     pi_bar = n_map - m_map * sp.I
94
95     coeff, factors_list = sp.factor_list(poly, extension=[sp.I])
96
97     psi_pi_factors = []
98     psi_bar_factors = []
99
100    for f, p in factors_list:
101        if is_Z_i_poly(f):
102            # If the factor is already in  $\mathbb{Z}[i]$ , split the power
              symmetrically
103            psi_pi_factors.append((f, p // 2 if p > 1 else 1))
104            psi_bar_factors.append((f, p // 2 if p > 1 else 1))
105        else:
106            # Test multiplying the monic fraction by pi and
              pi_bar

```

```

107         test_pi = sp.expand(f * pi)
108         test_bar = sp.expand(f * pi_bar)
109
110         # Whichever one clears the fractions dictates where
111         # the factor belongs!
112         if is_Z_i_poly(test_pi):
113             psi_pi_factors.append((test_pi, p))
114         elif is_Z_i_poly(test_bar):
115             psi_bar_factors.append((test_bar, p))
116         else:
117             psi_pi_factors.append((f, p)) # Fallback
118             safeguard
119
120     str_pi = format_grouped(psi_pi_factors)
121     str_bar = format_grouped(psi_bar_factors)
122
123     return f"{str_pi}{str_bar}"
124
125 # --- MATH LOGIC ---
126 def get_psi_basic(n, x, y):
127     psi_cache = {}
128     def psi(k):
129         if k in psi_cache: return psi_cache[k]
130         if k <= 0: return 0
131         if k == 1: return 1
132         if k == 2: return 2*y
133         if k == 3: return 3*x**4 + 6*x**2 - 1
134         if k == 4: return 4*y*(x**6 + 5*x**4 - 5*x**2 - 1)
135         m = k // 2
136         if k % 2 == 0:
137             res = (psi(m) * (psi(m+2)*psi(m-1)**2 - psi(m-2)*psi(
138                 m+1)**2)) / (2*y)
139         else:
140             res = psi(m+2)*psi(m)**3 - psi(m-1)*psi(m+1)**3
141         psi_cache[k] = sp.expand(res)
142         return psi_cache[k]
143     return psi(n)
144
145 def main():
146     print("\nDefine_the_Map_ _=_n_+_mi")
147     try:
148         n_input = input("Enter_n_(Integer):_")
149         n_map = int(n_input) if n_input else 0

```



```

149     m_input = input("Enter m (Complex): ")
150     m_map = int(m_input) if m_input else 0
151 except ValueError:
152     return
153
154 x, y = sp.symbols('x y')
155 curve_eq = x**3 + x
156
157 def clean_poly(expr):
158     return sp.expand(expr).subs(y**2, curve_eq).expand()
159
160 print(f"\n...Computing components for n={n_map}+{m_map}i...")
161
162 p_n = get_psi_basic(abs(n_map), x, y)
163 p_m = get_psi_basic(abs(m_map), x, y)
164
165 phi_n = clean_poly(x * p_n**2 - get_psi_basic(abs(n_map)+1, x, y) * get_psi_basic(abs(n_map)-1, x, y))
166 phi_m = clean_poly(x * p_m**2 - get_psi_basic(abs(m_map)+1, x, y) * get_psi_basic(abs(m_map)-1, x, y))
167
168 # Calculate Psi_pi
169 Psi_pi_final = clean_poly(phi_m * p_n**2 + phi_n * p_m**2)
170
171 print("\n" + "="*60)
172 print(f"RESULTS FOR n={n_map}+{m_map}i")
173 print("="*60)
174
175 print(f"\n  (Gaussian_Division_Denominator):")
176 print(format_poly_clean(Psi_pi_final))
177
178 print(f"\n  *_bar (Factorization):")
179 try:
180     print(format_factor_pair(Psi_pi_final, n_map, m_map))
181 except Exception as e:
182     print(f"Factorization_display_error:{e}")
183
184 if __name__ == "__main__":
185     main()

```

A.2 Computational Results

This section provides numerical verification for the algebraic properties of the Gaussian endomorphisms derived in Chapter 3. By utilizing the symbolic algebra implementation in Section A.1, we have calculated the division polynomials for several Gaussian primes π where

$$N(\pi) = p \equiv 1 \pmod{4}.$$

The primary goal of these computations is to demonstrate the factorization $\Psi_\pi = \psi_\pi \psi_{\bar{\pi}}$, verifying that the p -torsion subgroup indeed decomposes into two distinct eigenspaces of the endomorphism ring $\mathbb{Z}[i]$.

A.2.1 Factorization and Eisenstein Verification Table

The following table lists the irreducible Gaussian factors $\psi_\pi(x)$ for the curve $y^2 = x^3 + x$. All intermediate coefficients are checked for divisibility by π in $\mathbb{Z}[i]$ [cite: 524-527].

Prime π	$N(\pi)$	Irreducible Factor $\psi_\pi(x) \in \mathbb{Z}[i][x]$
$1 + 2i$	5	$(1 + 2i)x^2 + 1$
$1 - 2i$	5	$(1 - 2i)x^2 + 1$
$2 + 3i$	13	$(2 + 3i)x^6 + (4 - 7i)x^4 + (10 - 11i)x^2 - i$
$2 - 3i$	13	$(2 - 3i)x^6 + (4 + 7i)x^4 + (10 + 11i)x^2 + i$
$3 + 2i$	13	$(3 + 2i)x^6 - (7 + 4i)x^4 - (11 + 10i)x^2 - 1$
$3 - 2i$	13	$(3 - 2i)x^6 - (7 - 4i)x^4 - (11 - 10i)x^2 - 1$
$1 + 4i$	17	$(1 + 4i)x^8 + (20 + 12i)x^6 - (10 + 28i)x^4 - (12 + 20i)x^2 + 1$
$4 + i$	17	$(4 + i)x^8 + (12 + 20i)x^6 + (28 - 10i)x^4 + (20 - 12i)x^2 + i$
$2 + 5i$	29	$(2 + 5i)x^{14} + (76 - 13i)x^{12} + (246 + 325i)x^{10} + (520 + 459i)x^8 + (398 + 183i)x^6 - (148 + 65i)x^4 - (70 - i)x^2 + i$
$5 + 2i$	29	$((5 + 2i)x^{14} - (13 + 76i)x^{12} + (325 + 246i)x^{10} + (459 + 520i)x^8 + (183 + 398i)x^6 + (65 - 148i)x^4 - (1 - 70i)x^2 + 1)$

Figure A.1:

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